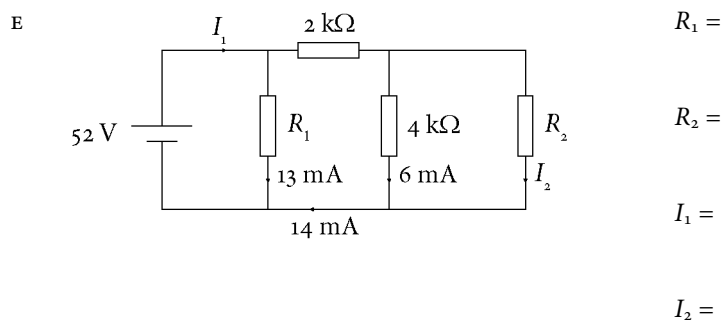
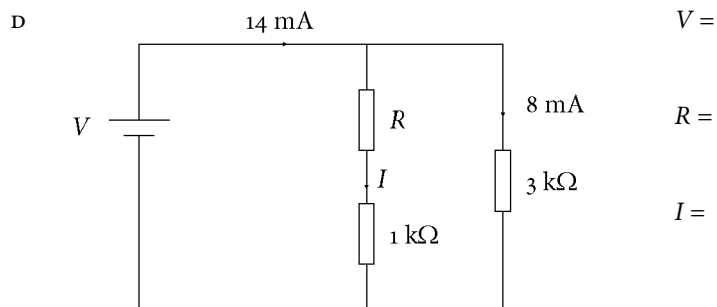
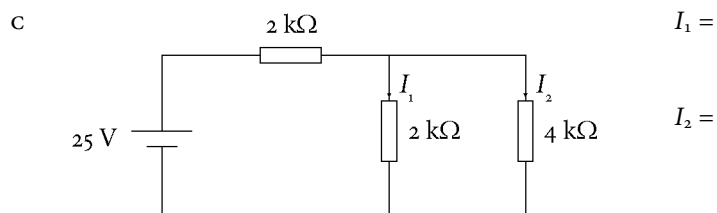
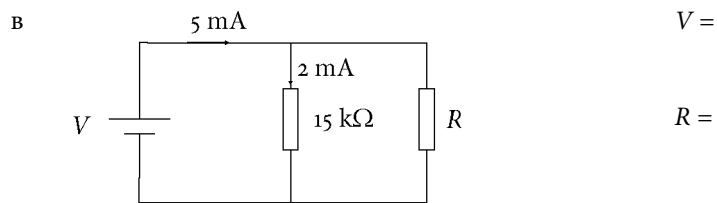
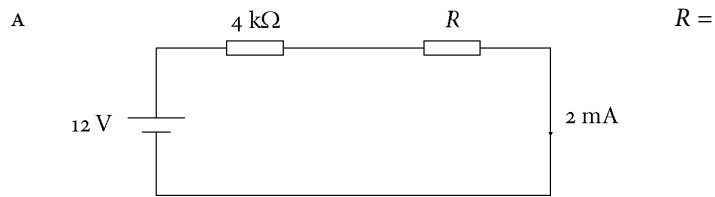


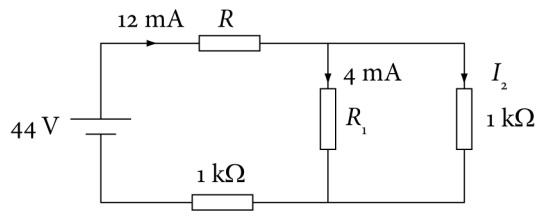
RESISTOR CIRCUITS

Applying the rules for voltage and current dividers, find the missing quantities in the five resistor circuits.

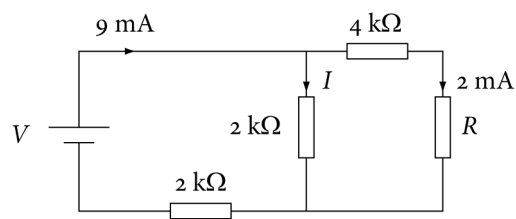


SOLUTIONS: A 2 kΩ; B 30 V, 10 kΩ; C 5 mA, 2.5 mA; D 24 V, 3 kΩ, 6 mA; E 4 kΩ, 3 kΩ, 27 mA, 8 mA

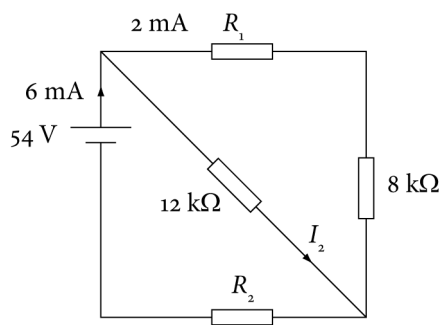
F

 $R =$ $R_1 =$ $I_2 =$

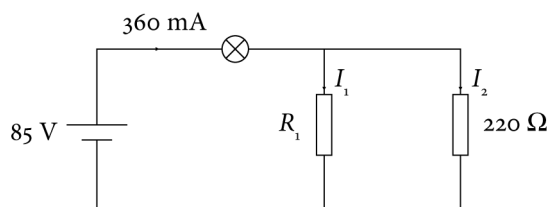
G

 $V =$ $R =$ $I =$

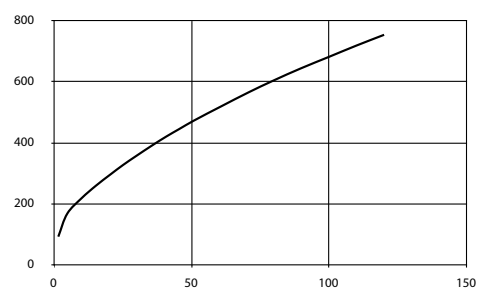
H

 $R_1 =$ $R_2 =$ $I_2 =$

I

 $R_1 =$ $I_1 =$ $I_2 =$

Characteristic of the light bulb:

 I [mA] vs. ΔV [V]

SOLUTIONS: F $2\text{ k}\Omega$, $2\text{ k}\Omega$, 8 mA ; G 32 V , $3\text{ k}\Omega$, 7 mA ; H $16\text{ k}\Omega$, $1\text{ k}\Omega$, 4 mA ; I $0.5\text{ k}\Omega$, 250 mA , 110 mA